Jargonaut

Evidence-Grounded Contract Simplification
Group: G1T5
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1. Background and Problem Statement

Individuals may sign contracts and agreements without fully reading or understanding them because these documents are lengthy and contain unfamiliar legal terminology. Even when users read the document, locating relevant information and interpreting its conditions can be difficult. These misunderstandings may initially appear harmless but can have significant consequences.
Addressing this problem requires both identifying the relevant contractual evidence and explaining it clearly. Returning an extracted clause alone may leave the user facing the same legal jargon that caused the confusion.

2. Proposed Solution

We propose a task-oriented chatbot that allows users to upload contracts or agreements and ask questions about their contents. The chatbot will retrieve relevant passages and answer in plain English.
Research Question: Does combining evidence extraction with a separately trained simplification component improve understandability while preserving contractual meaning?*
**The initial scope will cover selected topics in English commercial contracts.*

3. Dataset & Model Feasibility:

Datasets:

1. **CUAD:** 510 SEC contracts with 13,000+ expert annotations across 41 clause categories, including category questions, expected answer formats and gold evidence spans. **We use CUAD to train the evidence extractor to identify the contractual evidence relevant to a question.**

    - **Q&A Augmentation:** Plain-language questions and evidence-grounded answers are generated offline (using Claude/OpenAI) from CUAD annotations to better represent user-facing QA, while CUAD's expert evidence spans remain the extraction targets. Unrelated evidence is added as negative examples. **This makes the extractor more robust to natural user questions and unsupported queries.**

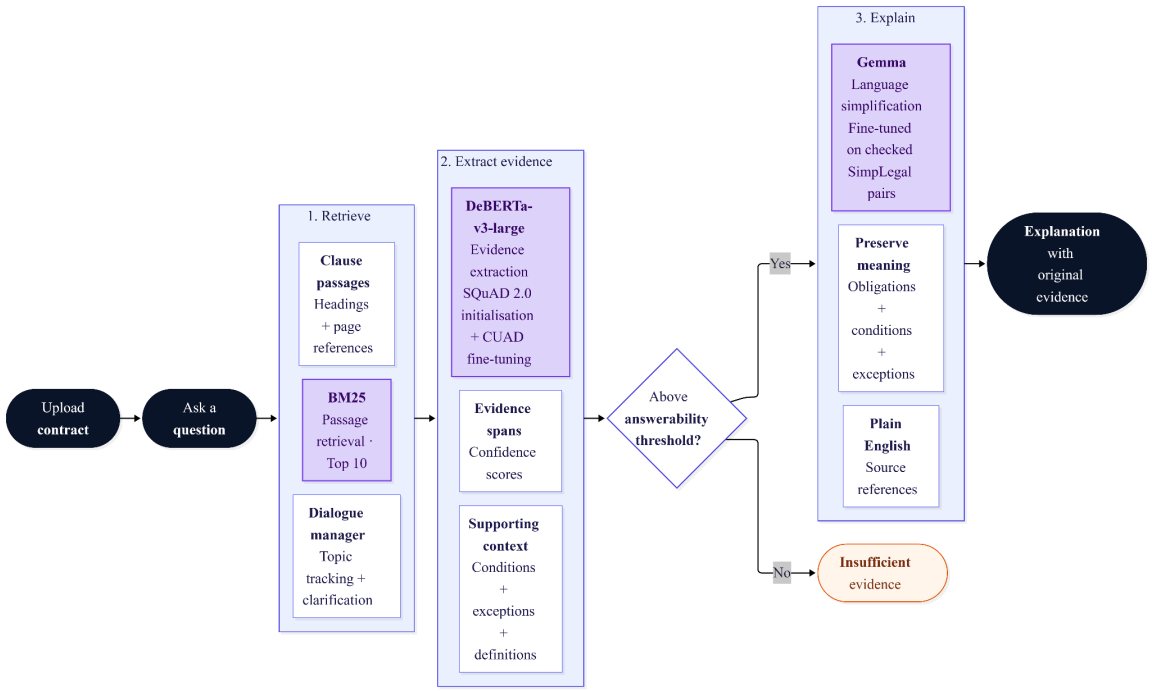
2**. SimpLegal-Pref:** 201 contract paragraphs rewritten at three reading levels. We retain the checked Basic rewrites (approximately Grade 8). **We use these legal-text → plain-English pairs to train Gemma to simplify extracted evidence while preserving its meaning.**

3. **ContractNLI**: 607 NDAs labelled against 17 hypotheses with supporting evidence sentences (423/61/123 split). **It supplements CUAD with confidentiality-related concepts and evidence supervision not covered by CUAD**, converted into question–evidence pairs for the extractor.

Model Choice:

1. **BM25:** Selected as a lightweight, interpretable lexical retriever; **Top-10 retrieval** provides broader coverage before evidence extraction.
2. **DeBERTa-v3-large:** Selected for strong extractive QA capability and SQuAD 2.0 initialization; **RoBERTa-base** serves as the simpler CUAD baseline.
3. **Gemma 4 31B with 4-bit QLoRA:** Selected for locally deployable generative simplification. We propose this with **gradient checkpointing** for parameter-efficient fine-tuning within a 24GB GPU budget, with **Gemma 4 E4B using LoRA** as a lower-resource fallback.

4. Architecture and User Flow



Stage 1: Document Processing and Retrieval

Contracts are split into overlapping, clause-based packages with headings and page references preserved for traceability. A **BM25 retriever** will identify top 10 candidate passages for each question. A lightweight dialogue tracks the active contract/topic for follow ups and requests clarification when needed.

Stage 2: Evidence Extraction

DeBERTa-v3-large, initialised from deepset's SQuAD 2.0 checkpoint, is fine-tuned on CUAD together with the augmented plain-language Q&A and negative examples described in Section 3, with ContractNLI extending confidentiality-related coverage. Given BM25's retrieved passages, the model predicts **answer spans and confidence scores**, retaining neighbouring conditions and referenced definitions where necessary. A validation-tuned answerability threshold determines whether evidence is sufficiently supported; otherwise, the system returns a fixed insufficient-evidence response.

A **RoBERTa-base CUAD model** serves as the baseline, with DeBERTa, plain-language augmentation, BM25 negatives and ContractNLI evaluated as successive extensions.

Stage 3: Legalese-to-English Conversion

The extracted evidence span and supporting context are passed to **Gemma for controlled simplification**. Gemma is fine-tuned using **LoRA/QLoRA on the checked SimpLegal-Pref Basic pairs**, learning a direct legal-text → plain-English transformation. The generated explanation should preserve parties, obligations, conditions, exceptions, negation, dates and quantities, and is displayed alongside the original evidence and source reference.

5. Evaluation Metrics

Model	Evaluation Metric	Reasoning
BM25	Recall@k	Missing the gold evidence means that DeBERTa will not be able to recover it, therefore, recall prioritised.
DeBERTa	Exact Match + F1 answer Precision/Recall	evidence extraction accuracy, checking for supported vs unsupported evidence, prioritising precision.
Gemma	Flesch–Kincaid + SARI + faithfulness check	simplification quality and preservation of legal meaning.

6. Project Contribution:

Member	Contribution
Kiara, Ridheema	-BM25 Retrieval & evaluation through Recall@K -DeBERTa model training & evaluation
Ian, Shernice	-Gemma simplification model training & evaluation
Theo	-Data preparation & augmentation -Frontend

All of us will take equal part in error analysis, testing and end-end integration.